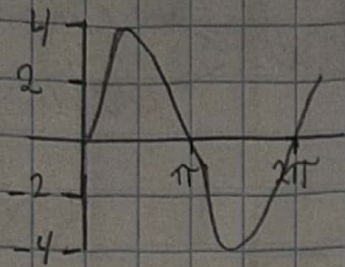


Tarea sección 5.5

1) a) $y = 4 \operatorname{sen} x$

$a = 4$

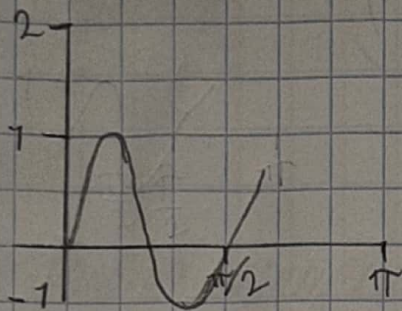
$p = 2\pi/1 = 2\pi$



b) $y = \operatorname{sen} 4x$

$a = 1$

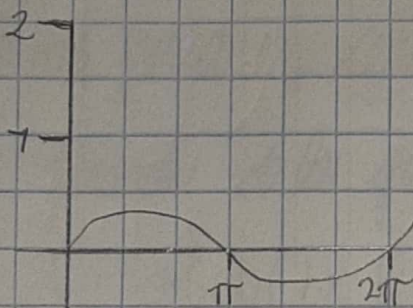
$p = 2\pi/4 = \pi/2$



c) $y = \frac{1}{4} \operatorname{sen} x$

$a = \frac{1}{4}$

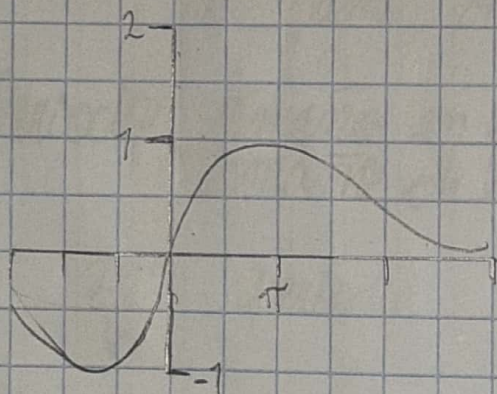
$p = 2\pi/1 = 2\pi$



d) $y = \operatorname{sen} \frac{1}{4}x$

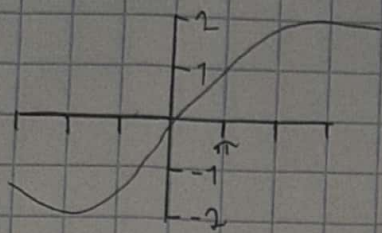
$a = 1$

$p = \frac{2\pi}{1/4} = 8\pi$



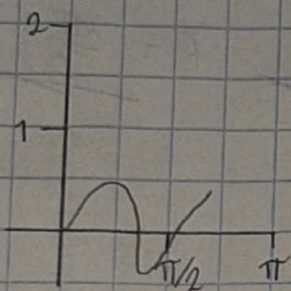
$$e) y = 2 \operatorname{sen} \frac{1}{4} x$$

$$a = 2 \quad p = \frac{2\pi}{1/4} = 8\pi$$



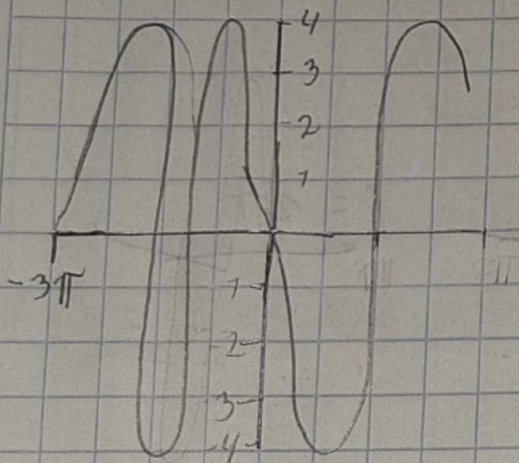
$$f) y = \frac{1}{2} \operatorname{sen} 4x$$

$$a = \frac{1}{2} \quad p = \frac{2\pi}{4} = \pi/2$$



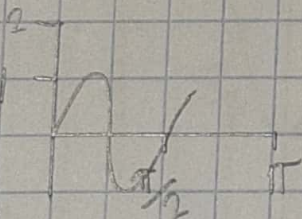
$$g) y = -4 \operatorname{sen} x$$

$$a = 4 \quad p = \frac{2\pi}{1} = 2\pi$$



$$h) y = \operatorname{sen} (-4x)$$

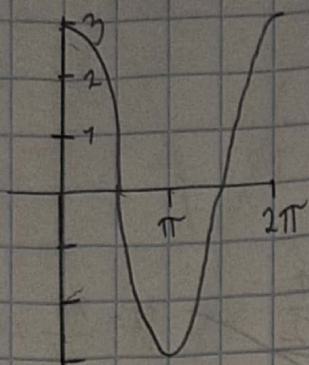
$$a = 1 \quad p = \frac{2\pi}{4} = \pi/2$$



3) a) $y = 3 \cos x$

$a = 3$

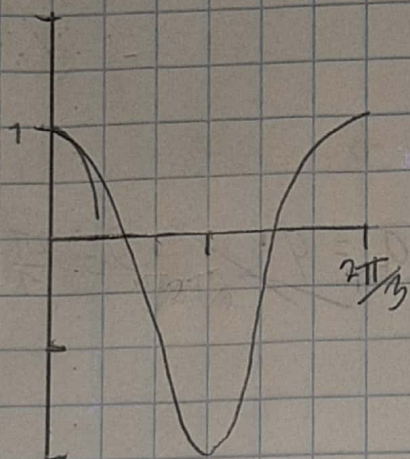
$p = 2\pi$



b) $y = \cos 3x$

$a = 1$

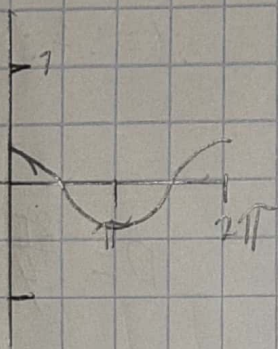
$p = \frac{2\pi}{3}$



c) $y = \frac{1}{3} \cos x$

$a = \frac{1}{3}$

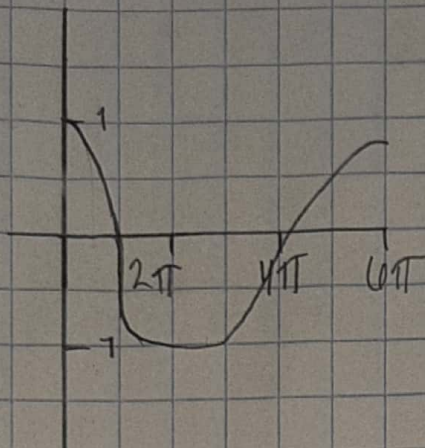
$p = 2\pi$



$$d) y = \cos \frac{1}{3}x$$

$$a=1$$

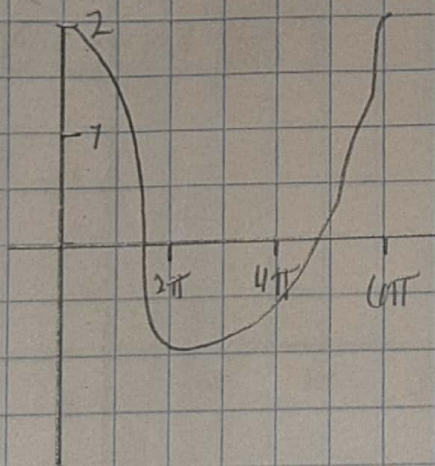
$$P = 2\pi / \frac{1}{3} = 6\pi$$



$$e) y = 2 \cos \frac{1}{3}x$$

$$a=2$$

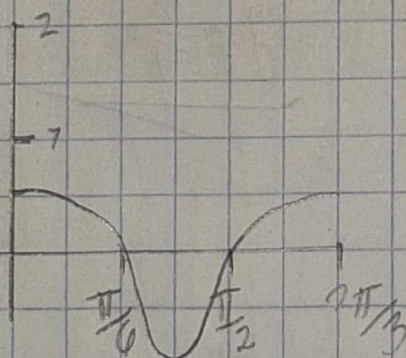
$$P = 2\pi / \frac{1}{3} = 6\pi$$



$$f) y = \frac{1}{2} \cos 3x$$

$$a = \frac{1}{2}$$

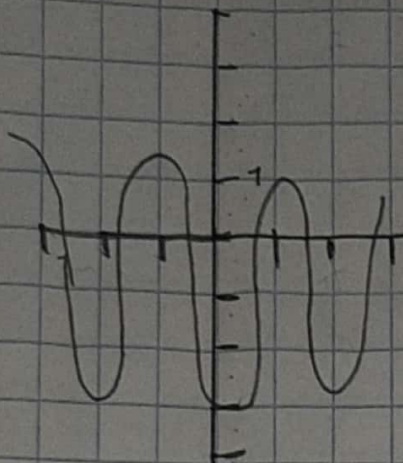
$$P = \frac{2\pi}{3}$$



$$g) y = -3 \cos x$$

$$a = 3$$

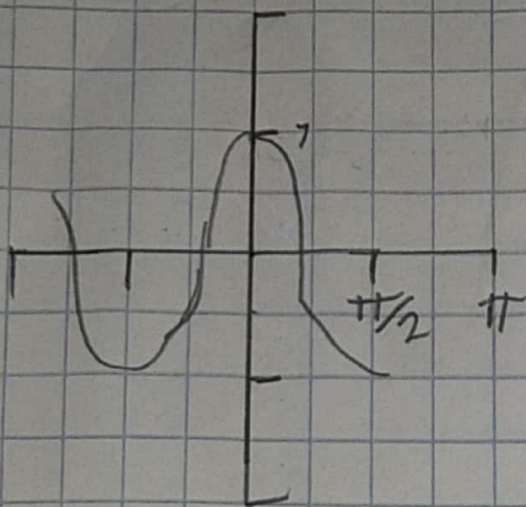
$$p = 2\pi$$



$$h) y = \cos(-3x)$$

$$a = 1$$

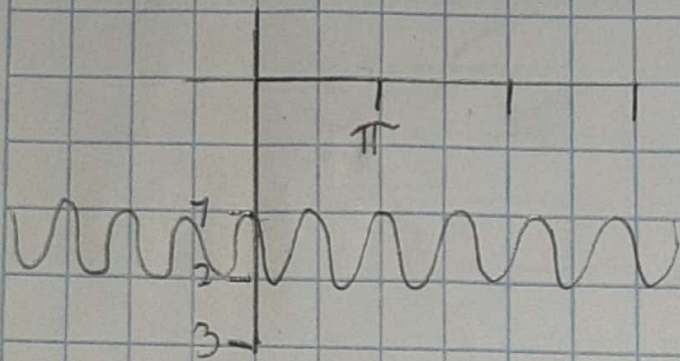
$$p = \frac{2\pi}{3}$$



$$k) y = -\cos(6x + \pi) - 2$$

$$a = 1 \quad p = \frac{2\pi}{6} = \frac{\pi}{3}$$

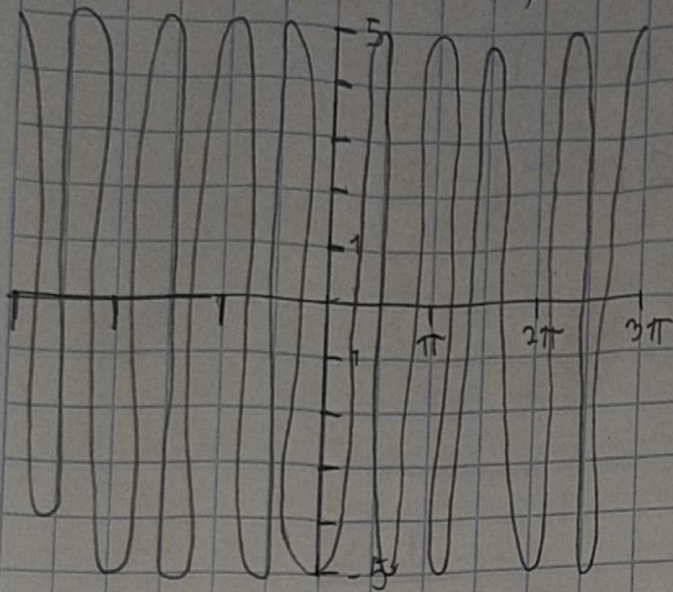
$$d = -\frac{\pi}{6}$$



$$27) y = 5 \sin(3x - \frac{\pi}{2})$$

$$a = 5 \quad \rho = \frac{2\pi}{3}$$

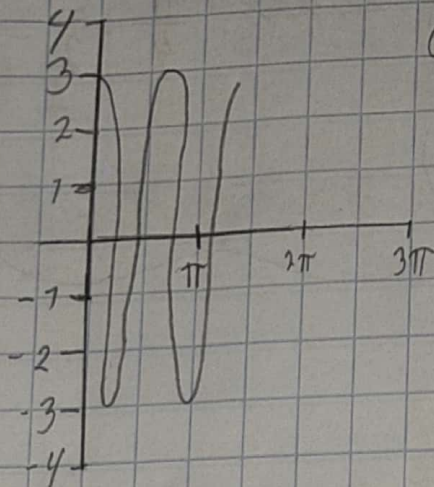
$$d = \frac{\frac{\pi}{2}}{\frac{2\pi}{3}} = \frac{\pi}{6}$$



$$33) y = 3 \cos(\pi x + 4\pi)$$

$$a = 3 \quad \rho = \frac{2\pi}{\pi} = 2$$

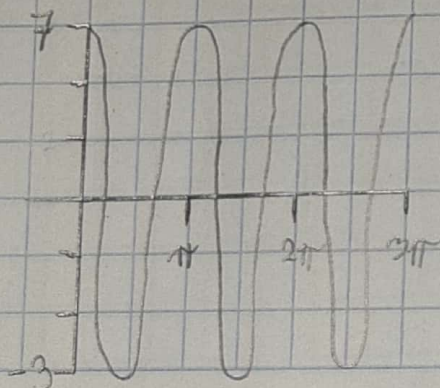
$$d = \frac{4\pi}{\pi} = 4$$



$$39) y = 5 \cos(2x + 2\pi) + 2$$

$$a = 5 \quad \rho = \frac{2\pi}{2} = \pi$$

$$d = \frac{2\pi}{2} = \pi$$



43) $y = 2 \sin\left(\frac{\pi}{2}x + \frac{3\pi}{2}\right)$

$a = 2$

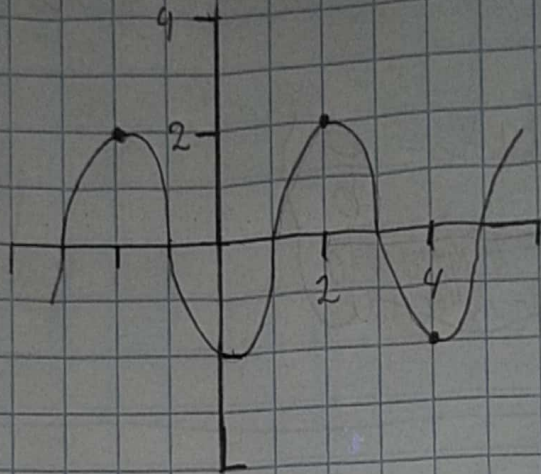
$b = 4$

$C = -\frac{3}{4} = \frac{-C}{2\pi}$

$C = \frac{6\pi}{4}$

$C = \frac{3\pi}{2}$

$b = \frac{2\pi}{4} = \frac{\pi}{2}$

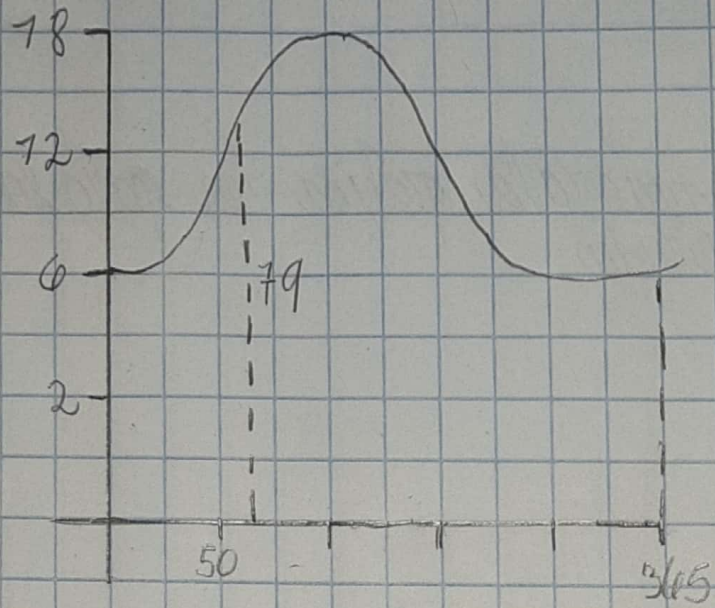


51) Horas de luz natural en Fairbanks

$D(t) = \frac{K}{2} \sin\left[\frac{2\pi}{365}(t - 79)\right] + 12$

$\frac{12}{2} = 6$ $D(t) = 6 \sin\left[\frac{2\pi}{365}(t - 79)\right] + 12$

$f(0) = 6 \sin\left[\frac{2\pi}{365}(-79)\right] = 6 \sin(-1.36) = -3.88$



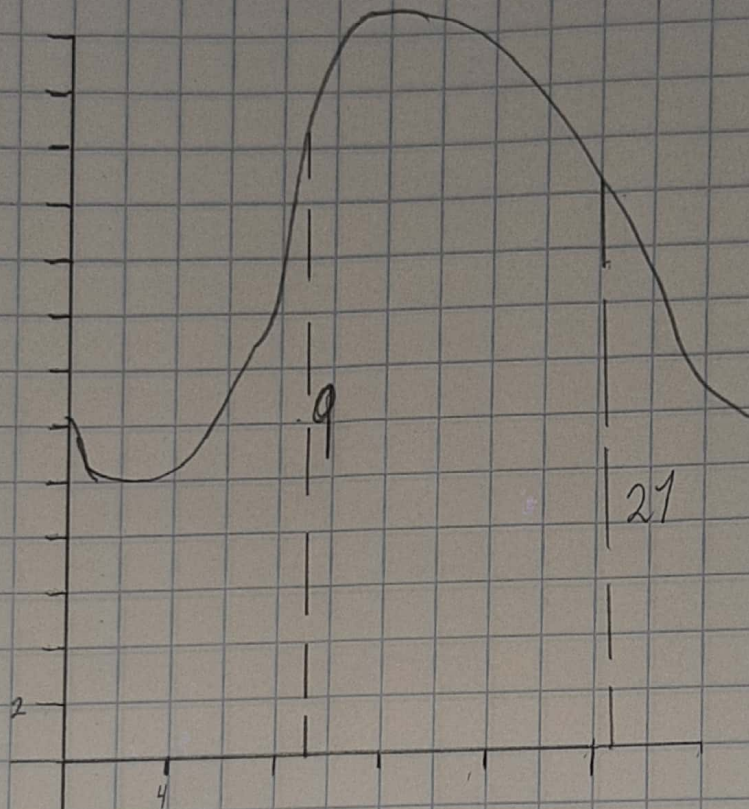
$$57) f(t) = 10 \operatorname{sen} \left[\frac{\pi}{12} (t - a) \right] + 20$$

$$a = 10$$

$$b = \frac{\pi}{12}$$

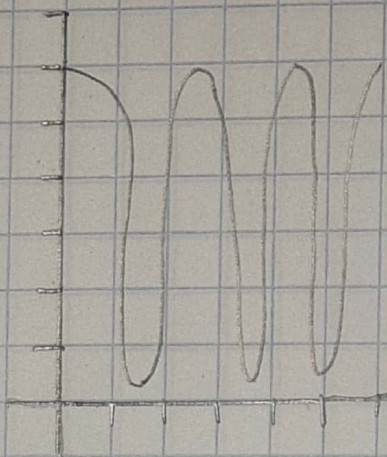
$$c = \frac{3\pi}{4}$$

$$d = 20$$



$$59) b) p(t) = 2.95 \operatorname{sen} \left(\frac{\pi}{6} t + \frac{\pi}{3} \right) + 3.75$$

a)



$$[0.5, 24.5, 5] \text{ per } [-7, 8]$$